



Technical Service Bulletin

Technical Service Bulletin: TSB120247

Released Date: 04-Sep-2013

New Piston and Piston Second Ring

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document announces the release of a new piston and piston second ring for use on QSK60 and QSK45 Tier 1 Diesel engines.

Note : This piston change does not apply to engines that use the low brake mean effective pressure piston. Only the ring pack change applies to engines using the low brake mean effective pressure piston.

The new piston and piston second ring have been released to improve engine durability.

The new piston has a cut back top land to prevent carbon buildup in this area.

The new piston second ring has a larger ring gap to reduce pressure buildup between the top and second rings during normal operation.

With the release of the new piston and second ring, a new piston and piston ring kit will be released. Present and new part numbers are shown in Table 1.

Present and new pistons and piston second rings can be mixed in an engine, but Cummins Inc. recommends the new piston and new piston second ring be installed together in the same cylinder.

Note : The present piston, piston second ring, piston kit, and piston ring kits, are still available for service on industrial and marine application engines.

This change went into effect on QSK60 Power Generation engines at the Daventry Engine Plant on 05-Sep-2012, with engine serial number (ESN) First 33194947.

The change was made on the Industrial and Marine engines at the Daventry Engine Plant on 08-Apr-2013. The engine serial number (ESN) First for Industrial engines are 33197388 (QSK60) and 33197652 (QSK45). The ESN First for QSK60 Marine engines is 33197451.

Table 1. New Part Numbers

Description	Present Part Number	New Part Number
Piston	3640474	3645958
Second Ring	3090460	3645611
Piston Kit	4955781	4309253
Piston Ring Kit	4089247	4309254

Use the following procedure in the QSK45 and QSK60 Service Manual, Bulletin 4021530, for ring gap measurements of the new piston second (intermediate) ring. Refer to Procedure 001-047 in Section 1. ([/qs3/pubsys2/xml/en/procedures/56/56-001-047-tr.html](#))

The new piston can be identified from the present one by the difference in the piston top land diameter (1). The new piston has a smaller diameter than the present one. Reference Table 2, for top land diameters of the new and present pistons (measured at the center of the top land).

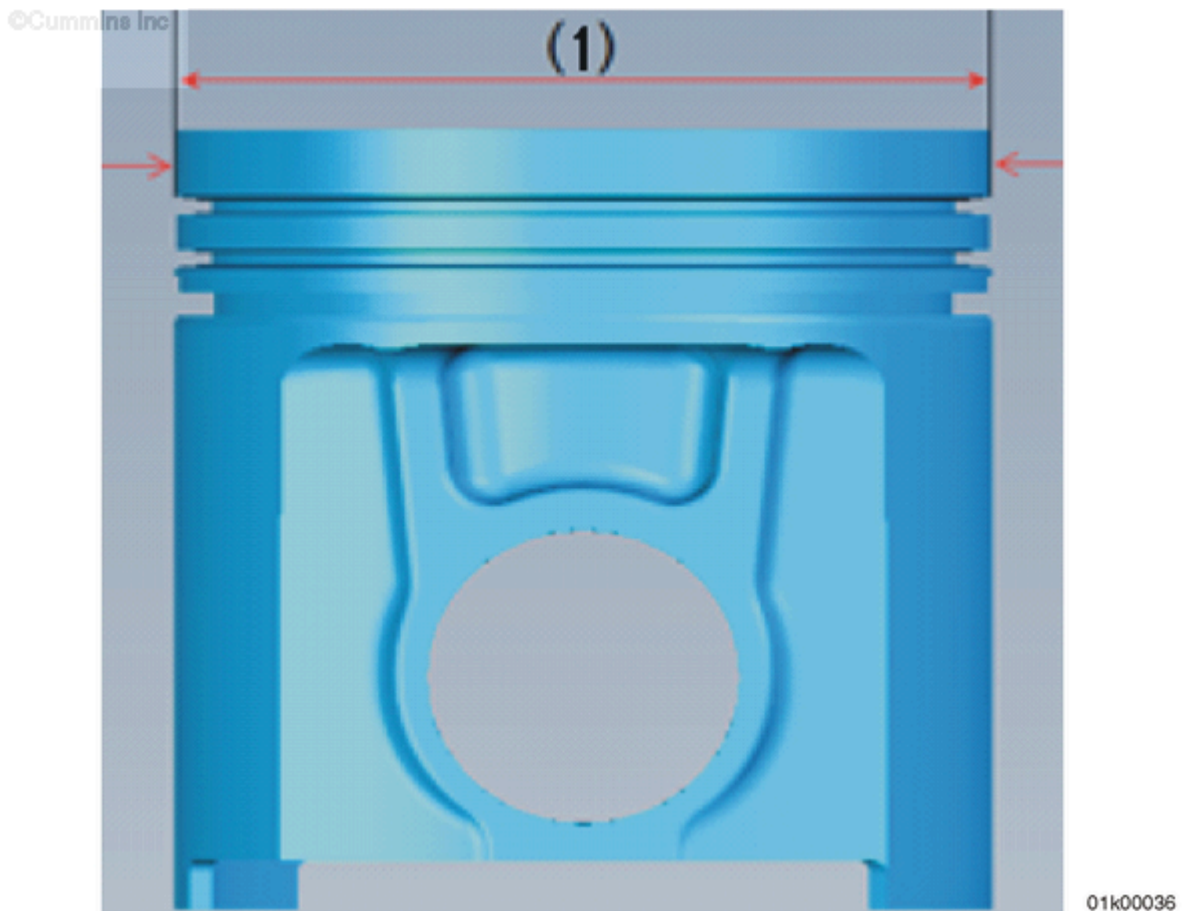
**Figure 1, Top Land Diameter Measurement**

Table 2: Top Land Dimensions				
Piston Part Number	Top Land Diameter (1)			
	Min		Max	
	mm	in	mm	in
3640474 (present)	156.91	6.178	157.01	6.181
3645958 (new)	155.71	6.130	155.81	6.134

Document History

Date	Details
2012-9-14	Module Created
2013-8-19	Updating information